Plant Identification Using Deep Learning and Image Processing

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***Abstract*—Species identification of plants is a key element of agriculture, conservation of biodiversity, botanical studies and environmental monitoring.But conventional techniques require expertise and time, particularly in cases where there is a resemblance in the appearance of different plant species.This paper presents a deep learning framework which integrates image processing and feature learning to automate the process of plant identification.To classify the images of plant leaves, the Leafsnap dataset is employed, and the background elements are eliminated using the open-source rembg tool, thereby reducing the noise caused by irrelevant information.The framework explores various deep learning models, such as ResNet-50, EfficientNetV2, ConvNeXt V2, Inception-based networks, Vision Transformer (ViT-B/16), and DINOv2-pretrained visual representations.A triplet based metric learning method is also integrated for further increasing the capability of the system to differentiate among the plant species that are similar in appearance.The models are evaluated according to their accuracy, precision, recall, F1-score, inference time and model size when the experiments are conducted under the same condition.The study also investigates the impact of background removal on the classification accuracy and considers performance vs. computational complexity.The aim is to propose a framework to compare convolutional networks, transformer-based models, and metric learning technique for automated plant recognition.The developed system can find applications in education, agriculture, botanical research and protecting environment.**

***Index Terms*—Plant identification, deep learning, image processing, Leafsnap dataset, background removal, convolutional neural networks, Vision Transformer, DINOv2, metric learning, triplet loss.**

I. INTRODUCTION

A. Background and Motivation

Plants play a vital role in the balance of the environment, provide a habitat for a diversity of life forms and provide vital resources like food, medicine and raw materials. Correctly identifying plant species is important in areas like farming, tree management, botanical studies, and environmental protection.The most common way to identify plants is for botanists to examine them based on their characteristics, such as shape, texture, vein patterns, color, and other physical features of the plant.This can be time consuming and complicated, particularly if the plants are similar in appearance.The problems reported here demonstrate the need for automatic systems that identify the plant species rapidly from digital images. The recent advances of artificial intelligence, image processing and Deep Learning have opened new opportunities to automatically identify plants. CNNs can learn complex visual patterns directly from images, and transformer-based models can capture the relationships between various aspects of an image.Such capabilities have spurned the development of deep learning as a promising approach to improve the speed and correctness with which plants can be classified.

B. Role of Image Processing and Deep Learning

The image pre-processing is one of the important stages in developing a reliable plant identification system. Important leaf features can be disturbed by background objects/scenes, shadows, changes in lighting and image noise.For this work, the background of the image is removed using the tool rembg to prepare the image for model training, to improve the focus of the model on the leaf surface.The Leafsnap dataset is the main source of data for training and testing the classification methods. Several deep learning models such as ResNet-50, EfficientNetV2, ConvNeXt V2, Inception-based models, and Vision Transformer-based models are explored. The advantages of learning features from the visual channel are also studied using a DINOv2-pretrained model.Moreover, a triplet loss-based metric learning paradigm is proposed to generate feature embeddings which can distinguish between visually similar plant species.

C. Research Problem and Objectives

Some progress has been made in classification of images, but it is still challenging to discriminate similar plants. Differences in leaf angle, shape, texture, and image quality can impact the accuracy of classification.Thus, for the effectiveness of these various model structures and feature-learning techniques a comprehensive comparison between them is required. The primary aim of this research is to design and evaluate a deep learning based approach for plant identification using image preprocessing and multiple plant classifiers. The goal of the research is to evaluate the effectiveness of the models on appropriate assessment metrics, study the influence of background removal, and investigate whether metric learning can

enhance the ability to discriminate plant species.The ultimate goal is to develop an education and agriculture friendly, rapid and accessible plant identification system for environmental management.

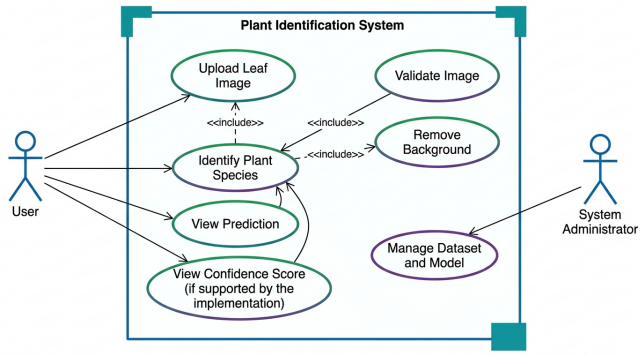


Fig. 1. Use-case diagram of the plant identification application.

II. LITERATURE REVIEW

A. Traditional Plant Identification and Deep Learning

The modern practice of plant identification based on the leaf image has been developed from manual plant feature extraction to automatic deep learning. Kumar et al. created a computer vision system called Leafsnap that is able to recognize tree species by leaf pictures and shape analysis [1].Their work established a benchmark for automated leaf recognition, but the recognition problem of differences between images captured in laboratory conditions and images captured outside of the lab is still challenging.Later, deep residual networks demonstrated the capability of directly learning distinguishing features from images, and deep learning has become a potential alternative to the traditional classification approaches [2].The influence of image conditions, the choice of model structure, and feature extraction on the recognition performance is still under investigation in recent studies [3,4].

B. CNN-Based Plant Species Classification

Convolutional neural networks (CNNs) are extensively adopted for plant classification as they can be trained to learn hierarchical visual patterns, like edges, veins, plant outlines, and textures. ResNet's residual connections allowed for simpler training of deeper networks [5] and Inception-based models were able to achieve better feature extraction by parallel convolutional operations [6].EfficientNetV2 emphasizes training efficiency and efficient use of parameters [7] and ConvNeXt V2 is a fusion of contemporary convolutional designs and self-supervised learning [8].These models offer a good basis for comparing the models across the various species of plants.

Comparative studies have also looked into using CNN ensembles and multistage classification techniques. BLeafNet includes CNN predictions and a fusion strategy [9] to enhance leaf classification for identification from leaf images.In this paper, SWP-LeafNET discusses a multi-stage deep learning approach to plant leaf recognition [10].In the year 2024, a research was conducted to analyze the performance of different deep learning models for the identification of medicinal plants, with varying results based on the specific model and set of data applied [11].

C. Indian Research on Medicinal Plant Identification

The medicinal plant identification research in India is of special interest because of the abundance of plants in the country and the long history of use of plants medicinally. Tiwari et al. had studied deep neural networks to classify various medicinal leaves [12].In [13] challenges faced were variations in lighting, projection and capturing distance, Ayur-PlantNet introduced a lightweight CNN approach for Indian Ayurvedic plants.Addition of more number of images of Indian medicinal plants acquired under different photographs in natural environment to the DIMPSAR dataset which will enable research on plant recognition and classification [14].In addition, studies have been conducted for real-time identification using deep learning and mobile apps [15].

These studies all highlight the requirement for systems which can be effective in real-world settings and which are capable of distinguishing between closely related species. However, the performance obtained from datasets of Indian medicinal plants can't be directly applied to Leafsnap because of variations in species distribution and image conditions.

D. Vision Transformers and Self-Supervised Learning

The vision transformer (ViT) views images as a sequence of patches and applies attention to capture the relationships between the various parts of an image [16]. DINOv2 improves self-supervised visual learning and offers pre-trained visual features for further classification applications [17].These strategies can be used to detect larger leaf structures that can be used along with the local structures learned by CNNs.

Another way of fine-grained recognition is by using metric learning. The triplet-loss methods learn embeddings by pushing the embedding of images of the same class together and pushing the embedding of images of different classes apart [18].This method works when there are two or more plant species with the same leaf shape or leaf texture.

E. Research Gap and Proposed Direction

The challenges that have been identified in recent surveys include variation in the datasets, differences in evaluation methods and the requirement for improved comparisons between learning techniques in plant image analysis [19] [20]. To address this, this study examines plant identification using Leafsnap with background removal using rembg and evaluates different models, such as ResNet-50, EfficientNetV2, ConvNeXt V2, Inception-based networks, ViT-B/16, DINOv2-pretrained features, and triplet-based metric learning models.The idea is that under the same experimental conditions, they should be able to assess how well they perform, instead of believing that any one model is the best.

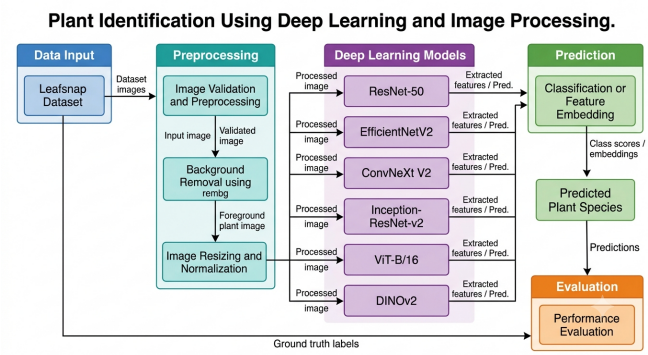


Fig. 2. Overall architecture of the proposed plant identification framework.

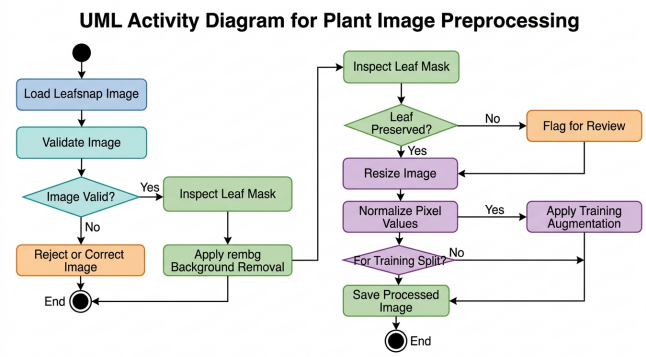


Fig. 3. Activity diagram of the plant image preprocessing pipeline.

III. METHODOLOGY

A. Dataset Collection and Preparation

Proposed plant identification system is based on the Leafsnap dataset which consists of leaf images collected in the laboratory and field [1]. The images are classified according to the plant species they depict.The data set is also verified for any damaged file, wrong labels, duplicate images and class imbalance before the training process.The data is divided into training, validation and testing sets to ensure the data are fair to compare.Great care is taken to ensure that images which are similar or duplicates do not occur in more than one set, in order not to bias the results.To guarantee a uniform image processing, all models go through the same steps.

B. Background Removal and Image Preprocessing

The open source rembg tool is used to remove the background of leaves to make them stand out. This step is useful for separating the area of the leaf from the rest of the image and helps to eliminate any other items from the image.The aim is to concentrate the model on the major features of the leaf, for example its shape, veins, edges and texture.Once preprocessed the images will be resized to fit the input size required by each model and will be modified based on the specific pre-processing done by each model.The model is enhanced by performing some changes to the training images such as rotation and flipping of the image and brightness adjustment.An assessment of the effectiveness of background removal is carried out by comparing the results of models trained on the original image with the processed image.

C. Deep Learning Model Development

The framework discusses in detail, different deep learning models for plant species identification. For learning complex patterns in images, ResNet-50 is chosen and for efficient use of computational resources, EfficientNetV2 is selected.ConvNeXt V2 is done due to its modern convolutional structure and enhanced feature learning [8].Multi-level feature extraction is believed to be performed by Inception-v4 or Inception-ResNet-v2 [6].In this work, Vision Transformer (ViT-B/16) is used to divide the image into smaller patches and interpret the relationship between the image parts based on self-attention mechanisms [16].Furthermore, DINOv2-pretrained models are evaluated on their usefulness of self-supervised learning for the task [17].If pre-trained weights are provided, then transfer learning is performed and trained for the specific task.

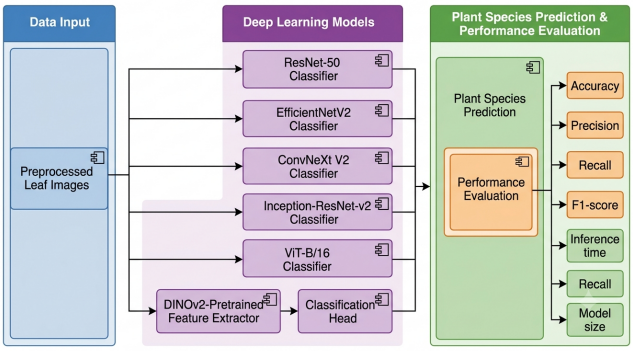


Fig. 4. Comparative architecture of the selected deep learning models.

D. Triplet-Based Metric Learning

The base model for designing an embedding network is an either ResNet-50 or EfficientNet model. The network is trained using triplet loss, which consists of an anchor image, a similar image in the same species and a different image in another species [18].The aim is to bring together the embeddings of images from the same species while pushing apart those from different species.The learned embeddings can be used later to classify images using nearest neighbour matching techniques or compared with traditional classification techniques to assess the effectiveness of the embeddings for discriminating between similar leaves.

E. Training and Model Evaluation

The same dataset splits and the same training settings are used for all models to ensure the models are trained fairly. Hyperparameters (learning rate, batch size, optimizer type, number of training rounds, and stopping criteria for early stopping) are noted.The accuracy, precision, recall and F1-score are used to measure the model performance.To come up with a list of plant species which are frequently confused, a

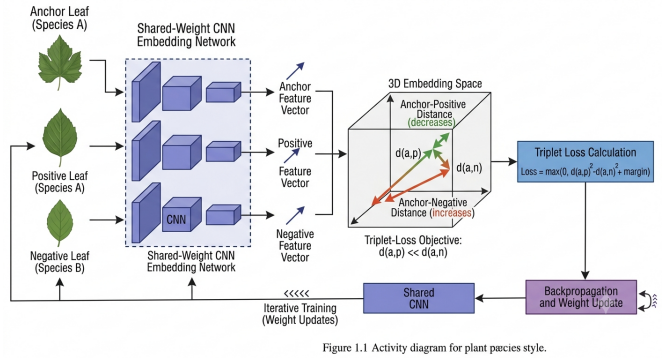


Figure 1.1 Activity diagram for plant species style.

Fig. 5. Activity diagram of triplet-based metric learning for plant identification.

confusion matrix is used.Assessing computational efficiency, the time required for the model to make predictions is also taken into account, along with the size of the model.The evaluation methods provide a comprehensive comparison of the different types of models in this study: convolutional networks, transformer-based models, and metric learning methods [3, 4, 9–15, 19, 20].

IV. EXPERIMENTAL SETUP

A. Experimental Environment

The plant identification system developed here has been constructed using Python, and is based on deep learning and image processing libraries. The Leafsnap dataset consists of images of plant leaves used for testing the system's ability to classify plant species using leaf images [1].Images are preprocessed by removing background using open source rembg tool prior to training the model.The setup can be used for training, validation, testing, and comparison of various models, including convolutional, transformer-based, and metric-learning models.The software used in the experiment should also be mentioned and the software and hardware used should be documented to make it possible to replicate the experiment.

B. Data set preparation and preprocessing

The data is organized by the name of the plant species and is divided into three sets: training, validation and testing. All models share the same division to make the results comparable.Images of each leaf are processed using rembg to remove any background, then resized according to the input requirement of each model, and then normalized by using the proper setting.The model is improved to generalise by using data augmentation methods such as brightness adjustments, flipping and rotation for training images only.Original images and background subtracted images can be evaluated independently to determine the impact of background subtraction.

C. Model Configuration

The models tested are ResNet-50, EfficientNetV2, ConvNeXt V2, Inception-based design, ViT-B/16 and DINOv2-pretrained model. These models include various techniques of feature extraction and classification [5]–[8], [16] and [17].Metric-learning model trained with ResNet-50 or EfficientNet is also used to investigate the system's ability to learn and differentiate between plant species using triplet loss [18].When available, pre-trained weights are used and the models are fine-tuned for the specific classification task.

D. Evaluation Strategy

The accuracy, precision, recall and F1 score are used to evaluate the performance of the models. A confusion matrix can be used to determine the common species that are misclassified.In addition, the time taken to make predictions and the size of the models are recorded to evaluate computational efficiency.This experimental set-up is based on the common procedure used for image classification of plants in the literature [3]–[15, 19]-[20].The model that works best in testing is chosen based on the results, rather than the assumptions.

TABLE I
EXPERIMENTAL CONFIGURATION

Parameter	Experimental Setting
Programming Language	Python
Dataset	Leafsnap
Background Removal	rembg
Preprocessing	Resizing, normalization, augmentation
CNN Models	ResNet-50, EfficientNetV2, ConvNeXt V2
Inception Architecture	Inception-v4 / Inception-ResNet-v2
Transformer Models	ViT-B/16, DINOv2-pretrained
Metric Learning	ResNet-50/EfficientNet with triplet loss
Data Split	Training, validation, and testing
Evaluation Metrics	Accuracy, precision, recall, F1-score
Additional Measurements	Inference time, model size
Deep Learning Framework	PyTorch or TensorFlow
Hardware	Actual CPU/GPU configuration

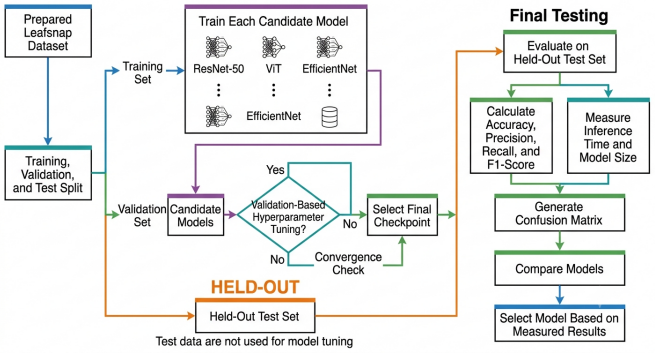


Fig. 6. Experimental evaluation and model selection workflow.

V. RESULTS AND DISCUSSION

A. Model Performance Analysis

The study, "Plant Identification Using Deep Learning and Image Processing," evaluates a number of deep learning models for plant species identification with the Leafsnap plant dataset. Comparisons are made with different architectures including ResNet-50, EfficientNetV2, ConvNeXt V2, Inception-based networks, ViT-B/16, DINOv2-pretrained features and a triplet-based metric-learning network.Previous research has demonstrated the effectiveness of deep learning in leaf classification, depending on the datasets, models and evaluation methods used [1, 3, 4]. For example, the level of accuracy achieved by DenseNet201 for medicinal plant identification in the public data set was reported as 99.64These results show that it's important to test models under different image conditions.However, the results cannot be simply applied to the present system due to the difference in the experimental conditions and data set.

B. Effect of Background Removal

A major operation to be carried out is the removal of the background using the open source rembg tool. This step is to reduce other elements and items around the leaf, which will make the shape, edges, veins and texture of the leaf easier to see.Previous studies on plant recognition also explored leaf segmentation and features extraction to aid in species recognition [1].Background removal, however, is not always associated with an improvement in accuracy.The segmentation may include part of the leaf margins or the important parts, which may reduce the model performance.Therefore, it is important to make comparisons between the original photographs and a set in which the background has been removed to see how effective this method is.

C. Comparison of CNN & Transformer Architectures

The selected CNN models are different in visual feature extraction techniques. ResNet-50 has residual connections to aid the process of deep learning, and EfficientNetV2 has been designed to be efficient in computation [5] [7].In the case of a modern convolutional design, as in ConvNeXt V2 [8], or multiple convolutional layers used to capture information, as in Inception-based models [6], the choice of model depends on the specific task.These models can be compared in terms of the classification accuracy, computational cost, and their power to discriminate between similar looking species. ViT-B/16 uses self-attention mechanisms to understand relationships between image patches [16], while DINOv2 uses self-supervised learning to create visual features [17]. These methods could be useful for recording general leaf features, but need to be tested.Multiple architectures, rather than a single best model for all plant datasets, appear to be a better approach [2, 11].

D. Evaluation of Metric Learning

The embedding network is based on the triplet structure, and investigates whether learning distances between leaf embeddings can aid in the separation of close species. The images of the same species are given similar embedding, and the images of different species are separated [18].This method could be especially helpful when species have similar leaf shapes or textures.The effectiveness should be compared with the traditional classification models with the same test set and suitable evaluation method [3],[4].

E. Overall Discussion & Limitations

The suggested framework provides a structured comparison of image preprocessing methods, CNN, Transformer network, and metric learning methods. Various metrics like accuracy, precision, recall, F1-score, inference time, and model size can be used to assess the accuracy of the predictions and the computational efficiency.Previous research also suggests that background complexity and differences between species can significantly impact classification results [9]–[15], [19], [20].

VI. CONCLUSION AND FUTURE SCOPE

A. Conclusion

In this paper, we present a deep learning-based framework that aims to recognize plant species based on image processing and visual feature learning. The Leafsnap dataset is the primary source of images, and background removal is done with the open source rembg tool to remove unwanted objects from the background and concentrate on the key features of the leaves.The framework explores several deep learning models, such as ResNet-50, EfficientNetV2, ConvNeXt V2, Inception-based networks, Vision Transformer (ViT-B/16), and DINOv2-pretrained models.Moreover, a metric-learning method based on triplets is incorporated for assessing the capability in discriminating plant species with similar visual appearances.

The different models are evaluated by their accuracy, precision, recall, f1 score, inference time, model size in the classification task. The research compares the performance of the convolutional networks, transformer-based models, and metric learning techniques in a similar setup to identify efficient techniques for automated plant recognition.Several applications of the system have been identified, such as educational, agricultural, biodiversity and environmental protection applications.But it is necessary to validate the performance of every model and the effect of background removal by experimental results.

B. Future Scope

This could be expanded in multiple ways in future development. More extensive and diverse data sets, with more plant species and greater variation in lighting, angles and backgrounds, could be used to test the framework.One of these avenues is to explore more sophisticated segmentation techniques that would enable the retention of more intricate leaf details while suppressing the background information.Other techniques such as model compression (e.g., quantization, pruning, knowledge distillation) might also be investigated to enhance efficiency and enable the model to be deployed on mobile devices.

When leaf images are inadequate for species identification, adding other plant attributes like flowers, fruit, stem, and bark may add to the ability to identify species. A user-friendly mobile or web application could enable users to submit plant images and get species predictions along with related botanical data.Using explainable AI techniques can also enable the visualization of how features affect the model's prediction, which can promote user trust.The upgrades might result in an easier to access, effective and scalable, plant identification system in real world scenarios.

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